

CHAPTER 2 : HEALTH AND DISEASES

Topic:-

Visit the public health centre or clinic nearby and collect information about vaccination.

Introduction:-

Our human body has a wonderful internal army called the immune system that defends us against invading disease germs. When harmful bacteria or viruses enter our bloodstream, white blood cells recognize them and produce specialized protective proteins called antibodies to destroy them. However, fighting strong germs for the first time takes time, during which a person can become severely ill or even die.

To protect children before dangerous germs attack, medical science developed the wonder of vaccination. A vaccine is a biological preparation containing weakened, dead, or inactive parts of a disease germ. When injected into a healthy body, it stimulates the immune system to produce antibodies and creates memory cells without making the child sick. In this project, we visited our local Primary Health Centre (PHC) to collect detailed information about vaccination.

❖ Key Vaccination Milestones in History:-

- **Edward Jenner (1796):** Discovered the world's first smallpox vaccine.
- **Louis Pasteur (1885):** Developed the lifesaving rabies vaccine for dog bites.
- **Pulse Polio Drive in India:** Successfully eradicated crippling polio from India in 2014!

Aim & PHC Visit Plan:-

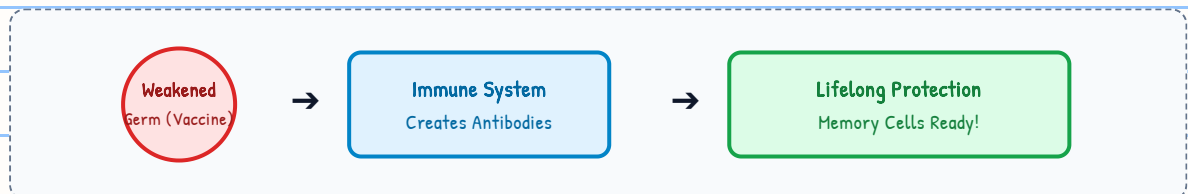
Aim: To visit the nearby government Primary Health Centre, interview the medical officer, and document the National Immunization Schedule for children and mothers.

❖ *Objectives of the Study:-*

- To record all vaccines administered from birth up to 16 years of age.
- To understand how the 'Cold Chain' keeps sensitive vaccines potent and safe.
- To learn how free vaccination programs protect public health in rural and urban India.

❖ *Materials Carried:-*

- Interview questionnaire notebook, pens, and school identity cards.
- Official Mother and Child Protection (MCP) immunization card copy.



Procedure & Medical Officer Interview:-

We visited the Primary Health Centre during Wednesday immunization hours and conducted our interview:

- **Q1: What vaccines are given immediately at birth?**

Doctor's Answer: Every newborn baby is given BCG (for tuberculosis), OPV zero dose (for polio), and Hepatitis B birth dose within 24 hours.

- **Q2: What is the Pentavalent vaccine?**

Doctor's Answer: It is a 5-in-1 combo vaccine protecting against Diphtheria, Pertussis (whooping cough), Tetanus, Hepatitis B, and Hib pneumonia, saving babies from multiple injections.

- **Q3: How are vaccines stored and transported?**

Doctor's Answer: Vaccines must be kept in special refrigerators between +2°C and +8°C. This temperature control is called the 'Cold Chain'. Vaccine carriers with blue ice packs are used during village visits.

- **Q4: Why are booster shots given at older ages?**

Doctor's Answer: Immunity can gradually decrease over years. Booster doses at 5, 10, and 16 years refresh memory cells.

National Immunization Schedule Table:-

The table below documents the official vaccination chart collected from the PHC:

<i>Age / Stage</i>	<i>Vaccine Name</i>	<i>Protects Against</i>	<i>Method of Delivery</i>
At Birth	BCG, OPV-0, Hep-B	Tuberculosis, Polio, Hepatitis B	Injection & Oral drops
6, 10, 14 Weeks	Pentavalent, Rota, PCV	Diphtheria, Tetanus, Diarrhoea, Pneumonia	Injections & Oral drops
9 to 12 Months	MR-1 (Measles-Rubella), JE	Measles, German Measles, Brain fever	Subcutaneous injection
16 to 24 Months	DPT Booster-1, MR-2	Diphtheria, Pertussis, Tetanus, Measles	Intramuscular injection
5 to 6 Years	DPT Booster-2	Diphtheria, Whooping cough, Tetanus	Intramuscular injection
10 & 16 Years	Td (Tetanus & adult diphtheria)	Tetanus and Diphtheria	Intramuscular injection

Scientific Analysis & Discussion:-

Analyzing the immunization schedule shows why vaccines are the most effective public health invention:

- **Mechanism of Immune Memory:** When a dead or weakened virus enters the body in a vaccine, special B-lymphocytes create antibodies. Even after the vaccine particles disappear, 'Memory Cells' stay alive in lymph nodes for decades. If the real wild germ ever attacks, memory cells produce millions of antibodies instantly to eliminate it before sickness starts.
- **Eradication of Lethal Epidemics:** Diseases like smallpox once killed millions. Through global vaccination, smallpox was completely eradicated from the world. Polio has been almost entirely wiped out in India.
- **Importance of the Cold Chain:** Vaccines contain delicate biological proteins. If the temperature exceeds 8°C, the vaccine breaks down and loses its ability to build immunity.

❖ Herd Immunity Concept:-

When over 90% of children in a town are vaccinated, the disease cannot spread, indirectly protecting tiny newborn infants who are too young for vaccines.



Everyday Applications & Government Schemes:-

The Indian government runs massive public health initiatives to ensure every child is vaccinated:

- **Mission Indradhanush:** A special flagship nationwide drive targeting remote tribal and rural areas where children missed routine shots.
- **Free Vaccines at PHCs:** All basic childhood vaccines are provided 100% free of cost at all government hospitals and municipal clinics.
- **ANM & ASHA Workers:** Dedicated rural community health nurses maintain village birth registers and visit houses to ensure zero dropouts.

❖ Removing Common Myths and Fears About Vaccines:-

- *Myth:* Vaccines cause high dangerous fever.
Fact: A mild fever for 1 day is normal and proves the body's immune system is responding well.
- *Myth:* Natural infection is better than a vaccine.
Fact: Natural diseases like polio or measles can cause permanent disability, blindness, or death!

❖ Parents' Golden Rule:-

Always preserve the Mother and Child Protection (MCP) card safely, as school admissions and travel visas require proof of completed vaccination.

What We Learned & Conclusion:-

- Vaccines use harmless weakened or killed microbes to train our body's immune system to produce antibodies.
- The National Immunization Schedule protects infants from over 10 fatal diseases from birth to adolescence.
- The cold chain system keeps vaccines at a steady 2°C to 8°C to maintain their potency and safety.
- Vaccination is a safe, painless, and completely free shield that protects every child and community.

Conclusion:-

Our visit to the Primary Health Centre showed us how medical science protects human society from fatal epidemics. Vaccines are truly life-saving shields developed by science. By ensuring every child receives all scheduled vaccines on time, we can build a strong, healthy, and disease-free nation.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 2: Health and Diseases).
- National Health Mission (NHM) Universal Immunization Programme (UIP) Guidelines.
- Interview and records from Primary Health Centre, Medical Officer In-Charge.